

Plant Phenomics, From Sensors to Knowledge

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Major improvements in crop yield are needed to keep pace with population growth and climate change. While plant breeding efforts have greatly benefited from advances in genomics, profiling the crop phenome (i.e., the structure and function of plants) associated with allelic variants and environments remains a major technical bottleneck. Here, we review the conceptual and technical challenges facing plant phenomics. We first discuss how, given plants' high levels of morphological plasticity, crop phenomics presents distinct challenges compared with studies in animals. Next, we present strategies for multi-scale phenomics, and describe how major improvements in imaging, sensor technologies and data analysis are now making high-throughput root, shoot, whole-plant and canopy phenomic studies possible. We then suggest that research in this area is entering a new stage of development, in which phenomic pipelines can help researchers transform large numbers of images and sensor data into knowledge, necessitating novel methods of data handling and modelling. Collectively, these innovations are helping accelerate the selection of the next generation of crops more sustainable and resilient to climate change, and whose benefits promise to scale from physiology to breeding and to deliver real world impact for ongoing global food security efforts.

Introduction

Genetic improvement of crop resilience to abiotic stresses and new pests arising from climate change is imperative to ensure future food security [1,2]. The increasing use of gene editing [3,4] and continued exploitation of natural genetic variability [5,6] provide invaluable opportunities for generating novel alleles and selecting natural sources of genetic variation for crop improvement [2]. This requires analysis of hundreds of lines grown under diverse environmental scenarios. While genotyping has reached this throughput at a relatively low cost through advances in DNA marker assays and sequencing technologies [7], equivalent improvements to generate high-throughput and valuable phenotypic information are urgently needed [8]. This is the object of the field of plant phenomics, which we define here as the development and application of the suite of tools and methods used for three major goals — capturing information on structure, function and performance of large numbers of plants, together with their environment; analysing, organizing and storing the resulting datasets; and developing models able to disentangle and simulate plant behaviour in a range of scenarios.

Over the past decade, plant phenomics has made impressive progress, developing novel sensors and imaging techniques for a wide range of traits, organs and situations [8–10]. However, data handling and processing remain major challenges when translating sensor information into knowledge. This Review focuses on that translation. We first discuss the reasons why the challenges differ between plant and animal phenomics, which are largely due to the strong interaction between environmental conditions and plant structure, function and metabolism. We suggest the need for a multi-scale strategy that links physiological mechanisms with plant performance across genotypes and environments from the molecular to field scales, based on a series of novel approaches and techniques. We

then discuss the challenges of studying these processes in the naturally fluctuating conditions in the field *versus* controlled environment conditions [11–14]. Finally, we suggest that phenomics is entering a new stage of development, necessitating novel methods for data handling, statistical approaches and modelling to connect and interpret the knowledge generated at different scales (see Box 1).

One Genotype, Many Phenotypes in Plants

Plant phenomics does not consist of solely associating a genotype to one phenotype in a given condition (e.g., in a controlled environment), but rather in characterizing the plasticity of the plant phenome when exposed to a range of environmental conditions. In contrast to most animals, which essentially retain the same structure regardless of their environment, a plant can form very different architectures depending on environmental conditions. For example, the same variety of *Arabidopsis thaliana* can exhibit a large 30-leaf plant or a small 8-leaf plant, after being exposed to either short or long day conditions, respectively (Figure 1A,B) [15]. Similarly, water deficit, nitrogen deprivation and low light have major effects on the number and size of plant organs (Figure 1B,C). As a consequence, plant phenomics research dedicates a large amount of effort to the study of variation in organism structure, whereas animal phenomics essentially focuses on metabolism.

Plant water status, temperature, fluxes or growth rate vary within minutes. Indeed, unlike mammals and birds, plants are not homeostatic for temperature and water under rapidly fluctuating environmental conditions. Plants transpire 50–200% of their own weight daily (vs. 1–2% for humans) [16], while their temperature follows their energy balance, resulting in rapid variations [17]. During a summer day, a plant can be at 11°C with a favorable water status in the early morning, but then experience 36°C and suffer severe

Box 1. Glossary of terms

Convolutional neural nets (CNNs): CNNs are a variant of traditional artificial neural networks (ANNs), machine-learning methods inspired by biological neuronal systems. Traditional neural nets take a pre-determined set of measurements or features as their input and learn to perform various tasks; classification is by far the most common. CNNs extend scope of ANNs, learning both how to achieve the task and what measurements are needed. CNNs operate over raw sensor data, and learn how to extract the necessary features.

Genome-wide association studies (GWAS): this consists in associating markers in the genome with phenotypes (omic, traits or yield) through a statistical analysis. The values of alleles at one genome position are associated with a quantitative increase or decrease of phenotypic values.

Genotype x environment interaction (GxE): the ranking of a set of genotypes differs between experiments for every trait or yield. GxE can be extracted from a statistical model, or can be analysed in detail using regressions of the considered trait with environmental variables. GxE is then analysed through the variability of slopes of these regressions.

Genomic selection (GS): represents a novel approach to marker-assisted breeding where, rather than attempting to identify individual loci significantly associated with a trait, GS uses all marker data as predictors of performance to deliver more accurate predictions.

Laser scanning systems: a 3D reconstruction method in which a known pattern of light (a line, grid or array of dots) is projected onto the target object by a laser light source. A camera, often fitted with a filter making the laser pattern easier to detect, views the reflected pattern. 3D is recovered from differences in the projected and viewed patterns of light.

Magnetic resonance imaging (MRI): MRI is a 3D imaging modality in which the target sample is placed in a strong magnetic field. Under these conditions some atomic nuclei, particularly hydrogen nuclei, absorb and emit radio frequency energy. Pulses of radio waves excite the hydrogen atoms, which emit signals that are detected by nearby antenna. The magnetic field allows these signals to be localised, mapping hydrogen atoms and so water.

Multi-view stereo: a 3D reconstruction technique in which multiple, usually colour, images are taken of a target object from different viewpoints. Features of interest are identified by independent analysis of each individual image. These features are then matched between images — features are matched if they are considered to depict the same point on the target object. The cameras' viewpoints are obtained by calibration and the 3D location of each object feature is recovered by triangulation.

Phenotype: here, we mean the profiling of the structures and functions associated with allelic variants, at the scales of cells (omic phenotyping), organs (main plant functions), whole plant (controls of these functions) and canopy (plant performance).

Quantitative trait loci (QTL): QTLs are regions of the genome containing one or more genes, associated with variation of a quantitative trait (phenotype). QTLs are identified by showing a statistical association between polymorphic markers and the measured phenotype.

Unmanned airborne vehicle (UAV): helicopters, drones or small planes able to fly over a field experiment, carrying a diversity of sensors. Their trajectory is programmed using GPS.

X-ray micro-computed tomography (μ CT): X-ray CT produces a 3D image in which each element (voxel) contains a value proportional to the density of the imaged object. The target object is placed on a rotating stage inside the imaging device. An emitter projects X-rays through the rotating sample to a detector on the other side of the device. The detector records the X-ray energy passing through the object. Density can be estimated from the difference in projected and detected X-ray energy.

water stress six hours later (Figure 1E), triggering spectacular changes in plant morphology (Figure 1D–F). Displacement transducers reveal that, under these conditions, plants exhibit rapid fluctuations in growth [18]. Leaf elongation can occur at a rate of 4 mm per hour at dawn *versus* 0 at 2pm [18]. Hence, although some degree of homeostasis exists at the cellular level [19,20], this is not the case at the organism level. Many molecular events occur during transitions between different environmental conditions [21], so phenomic analysis of non-stable states is essential. The low degree of homeostasis in plants also results in large functional consequences of the spatial variability of conditions a plant is exposed to [17]. For example, root system architecture exhibits large spatial variation reflecting local adaptation to highly heterogeneous soil water content [22,23]. Hence, the analysis of phenotypic datasets needs consistent time course information on environmental conditions as sensed by plants and organs, together with growth and physiology-related processes.

Because there is no central 'orchestrator' organ in plants [24], the control of most functions relies on feedback loops involving

different organs that exchange information through, for example, hormonal or hydraulic messages [25–27]. Such exchanges of information operate at short-term scales at the cell or organ levels, and translate into long-term plant or canopy behaviours through whole-plant mechanisms that are highly non-linear. Hence, plant phenomics requires analyses at spatial scales ranging from single cell to canopy, and temporal scales ranging from minutes (for metabolism and hydraulics) to months (for yield) (Table 1). Modelling is, therefore, an intrinsic part of plant phenomics, aimed at connecting these scales. Indeed, while often non-intuitive, feedback mechanisms are predictable using mathematical models [28].

Analysing the Plant Phenome across Spatial and Temporal Scales

Given the issues raised above, plant phenomics needs to capture and interpret a multi-dimensional matrix of functional and architectural variables measured at different scales (organ, plant, canopy), developmental stages and environmental

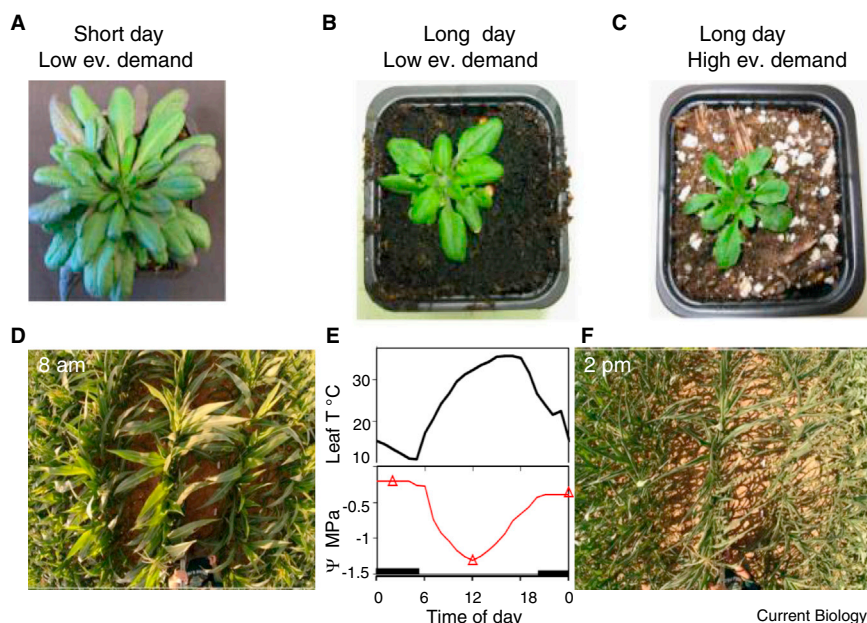


Figure 1. Illustrations of phenotypic plasticity.

Arabidopsis plants under low evaporative demand with short (A) or long (B) day, or under high evaporative demand (C) from [15]. Note the differences in leaf number and leaf size. (D,E) Maize plants in the morning and early afternoon and time courses of leaf temperature (T, from 11 to 36°C) and leaf water potential (Ψ, MPa) during the day. A leaf water potential of 0 MPa means free water, whereas −1.5 MPa is close to lethal values in many species. In the lower panel of E, symbols are measurements, lines are an interpolation using a model. In (F), the change in canopy aspect is due to leaf rolling, a symptom of water stress. Panels (D), (F) kindly provided by C. Fournier, INRA LEPSE Montpellier France.

scenarios. To address this inherent complexity, researchers have developed three categories of phenotyping platforms that have distinct objectives and employ different approaches and methods (Table 1).

High-Precision Platforms

These platforms operate at the organ level, most often over short time scales (Table 1). They aim at the identification of physiological mechanisms allowing plants to respond to changes in environmental conditions, and lead to the elucidation of their genetic control.

Profiling organ growth and architecture is used in high-precision platforms to uncover adaptive mechanisms associated with environmental signals. For example, X-ray micro-computed tomography (μCT; Figure 2A–D) of roots growing in soil macropores revealed the importance of direct contact with soil water to determine where new lateral root branches are positioned (Figure 2E) [23]. This translates into improved water and nutrient uptake (Figure 2F). Similarly, time-lapse 3D imaging of leaves combined with computational modelling allows identification of where and when tissue expansion and cell division occur [29]. In the case of leaves or sepals, this approach has revealed how new buds with few cells result in reproducible shapes through feedback between patterns of oriented growth and tissue deformation (Figure 3A) [30,31]. Analyzing the leaf elongation rate of maize plants with displacement transducers at high temporal resolution (i.e., minutes) in contrasting and fluctuating conditions allowed identification of a novel mechanism of drought adaptation. This mechanism involves regulatory interactions between circadian control of plant hydraulic properties, daily time course of evaporative demand and hydraulic properties of the rhizosphere (i.e., roots and the adjacent soil) [32].

The composition of plant tissues and the metabolite fluxes through organs can be characterized in ‘omics-based’ platforms for thousands of plants [33]. For instance Ionomics employs ICP-MS to perform elemental profiling [34]. This has allowed identification of *Arabidopsis* mutants whose leaves have altered

elemental composition. Many of these mutants were later shown to be defective for an impermeable barrier in roots, termed the Casparian strip, that regulates loading of elements into vascular tissues [35]. Metabolomic-based methods profile the compounds involved in major metabolic pathways. This approach has helped discover how different genotypes cope with environmental cues, uncovering the dialogue between the circadian clock and changes in light availability that allow plants to optimize the use of starch reserves [36]. Fluxomics (i.e., *in situ* imaging of the concentration and fluxes of elements [37]) have provided important insights about where, when and how water and nutrients are transferred in the plant [38]. For example, MRI-PET-based imaging has enabled researchers to map carbon flow from leaves to individual roots [39]. Imaging-based phenomic approaches can also be employed in cell-scale profiling studies. For example, eGFP- and FRET-based sensors have proved highly effective for monitoring the spatio-temporal dynamics of hormones and elements like zinc in *Arabidopsis* root cells [40–42] (Figure 3D) and uncover new mechanistic insights into their homeostatic regulation.

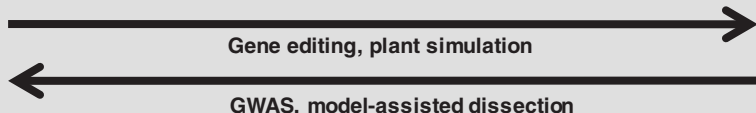
The examples above highlight how high-precision platforms are effective for discovering new physiological mechanisms, and also for upscaling them from organ to plant level. Nevertheless, at their current stage of development, these platforms cannot analyse the many thousands of plants needed to perform genetic studies across a range of environments over a whole life-cycle timescale. Hence, they are not directly relevant for upscaling mechanisms to predict important traits such as yield (Table 1).

Field Multi-Environment Networks

At the other extreme of plant phenomics, ‘field multi-environment networks’ (Table 1) are series of field experiments distributed in a geographical region, aimed at uncovering the genetics of yield stability. They probe the genetic control of plant performance in a range of environmental scenarios, without pre-conceived reference to a particular mechanism.

The yield of a given genotype often differs between field sites, as does the ranking of genotypes (genotype by environment interactions; G×E) [43,44]. Indeed, the relationship between yield and environmental conditions results from trade-offs between mechanisms that have distinct optima [45]. The relationship

Table 1. Phenotyping at different scales of organization.

Typical phenotyping platforms	High-precision platforms (omic, anatomy, organ)	Whole-plant, multi-environment platforms (field or controlled)	Field multi-environment networks
Level of plant organization	Organ	Plant or canopy	Canopies in a range of environments
Typical methods	Omics, 4D organ imaging, fluxes	4D plant/canopy imaging fluxes, sensors	Yield, sensor network, remote sensing
Typical mechanism	Hydraulics, metabolism, signalling	Light interception, water transfer, whole-plant signalling	Trade-offs between processes
Ratio biology/other processes			
Relevance for yield prediction			
Methods for trans-scale communication			

between genotype and phenotype therefore needs to be analysed in clusters of microclimatic conditions referred to hereafter as environmental scenarios [46,47]. For example, a network of 29 field experiments across Europe was used to grow a maize diversity panel and identify genomic regions associated with yield (quantitative trait loci, QTLs) under heat or water stresses [48]. Nearly all QTLs had conditional effects, positive, negative or null, depending on environmental scenarios. For instance, an allele at one QTL that controls the biosynthesis of the stress hormone abscisic acid (ABA) was favourable in drought but detrimental in well-watered situations. Hence, a large number (typically 20–40) of experiments needs to be conducted under diverse environmental conditions to explore such allelic effects. Genomic selection (GS) extends the former approach to establish predictions of the best combinations of alleles for yield [49]. GS requires phenotyping of hundreds/thousands of genotypes (the ‘training population’), in some cases together with the effects of environmental conditions [50]. The best combinations of alleles are then used to select, *in silico*, tens of thousands of plants, thereby avoiding direct phenotyping of these plants [50].

Whole-Plant, Multi-Environment Platforms

Given the complex interactions between QTL and environment and between QTLs [51], the interpretation of results generated by networks of field experiments is typically challenging, making it difficult to relate gene alleles with physiological mechanisms. To achieve this goal, a third category of plant phenomic platforms, ‘whole-plant, multi-environment platforms’, has been developed. These platforms are highly instrumented greenhouses or fields allowing one to follow and dissect variables such as the growth or transpiration of thousands of plants or small canopies, thereby allowing their genetic analysis (Table 1).

Highly automated platforms in greenhouses enable researchers to perform 4D characterisation of the architecture of shoot, root or canopy systems of hundreds of genotypes (Figure 3B,E; Figure 4A). They allow genetic analyses of traits such as root and shoot topology, angles, branching and growth rate as a

function of environmental conditions [52–56]. More elaborate traits such as the utilisation efficiency of water, light or nutrients can be calculated from these data using functional/structural plant models (Figure 2F; Figure 4J) [57–59]. This is illustrated in Figure 4, in which 4D imaging of whole plants and the mapping of incident light in a greenhouse makes it possible to disentangle the biomass accumulation of thousands of plants into well-defined processes, such as the amount of light intercepted by each plant (a function of leaf area and geometry) and the photosynthetic ability of each plant [58]. Crop models can then be used to connect the genetic variability of these processes to yield [60].

High-throughput field phenotyping has progressed rapidly in the last five years, based on the use of multi-spectral 4D analyses with sensors mounted on mobile systems such as gantries [61,62], ground vehicles or drones [63,64] (Figure 3C,F). They offer the possibility of estimating the genetic variability of yield, biomass accumulation and underlying processes in a variety of environmental scenarios. For example, canopy temperature provides a proxy for genetic differences in transpiration, which is often due to variation in root system architecture [63,64] (Figure 2F).

Cross-Scale Meta-Analyses

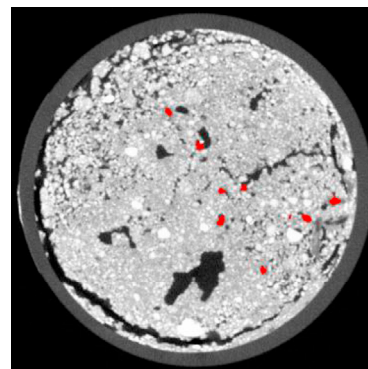
Currently, joint analyses of field experiments have been performed across years and sites [43,65]. While cross-scale approaches are also beginning to appear [66], they need to be developed further.

No single plant phenomic platform can analyse every scale, throughput or environment. For example, it would be misleading to measure yield in greenhouse experiments, as the amount and spectrum of light available to plants in a greenhouse and the distribution of roots in the soil in pots would make any attempt irrelevant [11]. Reciprocally, phenotyping of thousands of varieties in tens of field experiments is not compatible with costly and labour-intensive methods. A combination of approaches is therefore necessary, which we term ‘cross scale meta-analyses’. For instance, the plasticity of yield can be analysed in ‘field multi-environment networks’. The underlying genetic variability of trait adaptation can then be analysed in ‘whole-plant, multi-environment’ platforms for the same panels of genotypes, thereby associating

A X-ray CT phenotyping platform



B CT cross-section data



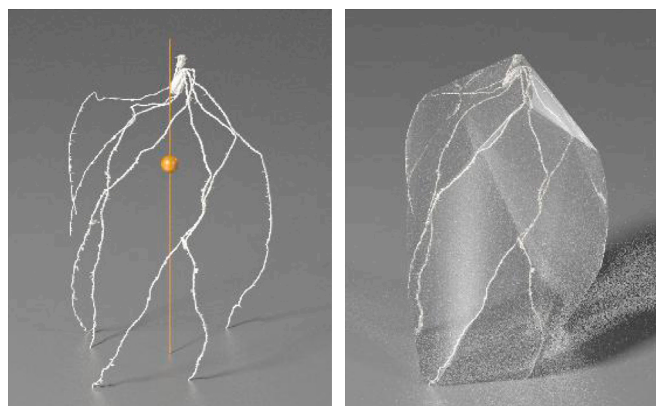
Roots in soil; 25 μ m resolution

C Root system recovery



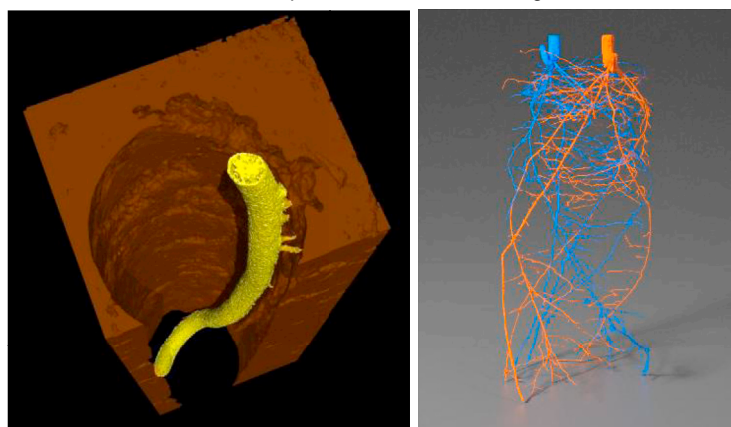
Root system of maize

D Quantification of root traits



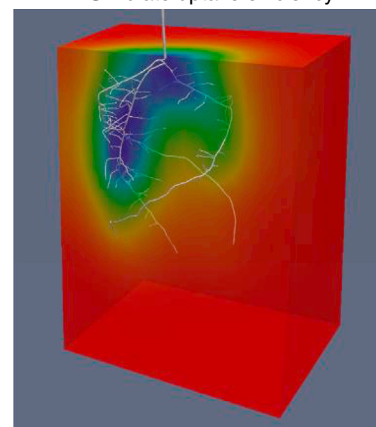
Quantification of global and local root system traits for plant phenotyping

E Response to environmental signals



Soil-root and root-root interaction

F Simulate uptake efficiency



CT data embedded in a finite element mesh

Current Biology

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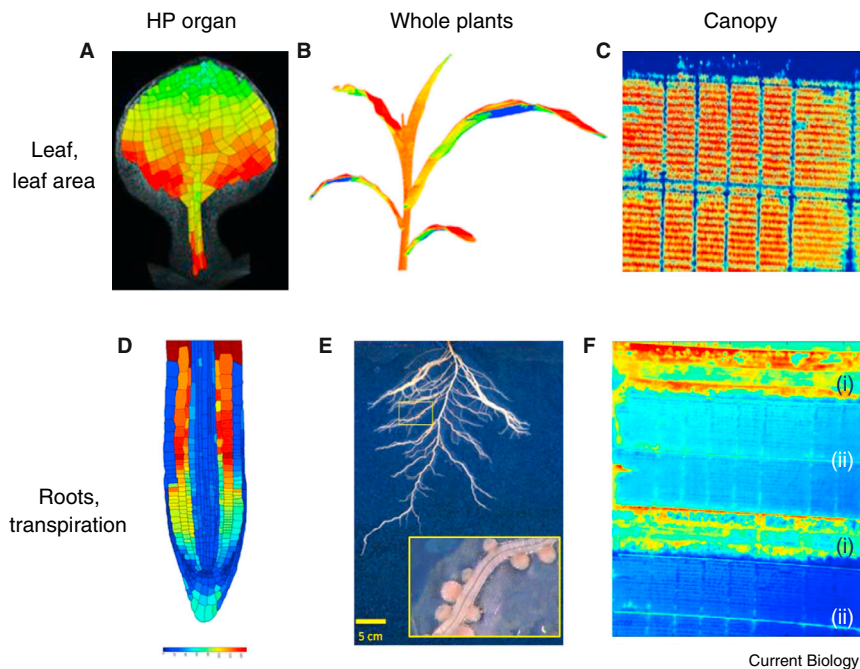


Figure 3. Novel imaging techniques at organ scale with high-precision (HP) platforms, at plant scale with whole-plant multi-environment platforms and at canopy scale.

(A) Heat map denoting areal leaf expansion rate using time-lapse imaging and computer modelling (red to green, rapid to slow growth), from [30]. Reprinted with permission from AAAS. (B) 3D representation of a maize plant from multiple images, at a throughput of 1000s plants/day. Colors indicate the amount of light received by each pixel of plant. (C) Multi-spectral (NDVI) image of a canopy; increasingly red colors represent increasing leaf area per unit m^2 of soil. (D) Image of an auxin biosensor in the *Arabidopsis* primary root obtained by confocal imaging [122]. (E) Whole-plant root system imaged in a rhizotron at throughput of 1000s plants/day. Inset, zoom on root nodules. From [53]. (F) Image of a canopy in the thermal infrared; increasingly red colors indicate lower transpiration rate, often linked to an unfavorable root system. Horizontal regions with distinct colors: (i) non-irrigated plot, (ii) irrigated plot. Note in (i) the superposition of spatial patterns with specific effects of genotypes in different plots. Panel B kindly provided by C. Fournier, INRA LEPSE Montpellier France. Panels (C) and (F) kindly provided by F. Baret, INRA CAPTE Avignon France.

Current Biology

QTLs affecting yield in specific environments to allelic variations of traits. The resulting alleles can be tested for their effects on mechanisms of plant adaptation in ‘high precision’ platforms [67]. Such meta-analyses are particularly vital in the case of root studies, in which the root architecture or growth can only be analysed in ‘high precision’ using ‘whole plant multi-environment platforms’, whereas only consequences can be observed in the field, for instance through differences in canopy temperature (Figure 3D–F).

Employing Trans-Scale Analyses to Link Sensors with Knowledge

Cross-scale meta-analyses, as defined above, require consistent methods for recovering data across all platforms, time scales and levels of plant organization (Table 1 and Figure 3). We discuss below the major challenges researchers face to achieve this ambitious, yet essential, next step in plant phenomic research.

Environmental Characterization, Sensor Networks

In our own experience, the analysis of datasets originating from different experiments and groups often suffers from a lack of consistent environmental information, which makes it impossible to analyse and model the differences in plant behaviour between experiments. To that end, several research consortia have proposed ‘minimum environmental datasets’ with the necessary environmental variables and protocols for data analysis and modelling at any scale [68,69]. Furthermore, a full environmental characterisation is now being facilitated by rapid progress in sensor technology. Cost-effective sensors can now be placed

in wireless networks to characterize the micro-environment of many organs in a plant and many plants in a canopy (Figure 5, arrows 1 and 2). This progressively applies to the characterization of the soil environment by combination of soil sensors with modelling [70]. This local information can be scaled up to whole-plant, field or regional levels using local, UAV or satellite imaging, respectively. This allows efficient mapping of environmental variables, thereby characterizing and capturing the effects of the spatial and temporal variation of growth conditions sensed by individual plants or fields (Table 1).

Consistent Analysis of Images and Time Series

Imaging systems have progressed exponentially in recent years, with a variety of non-invasive and information-rich techniques (e.g., laser microscopy and rangefinders, X-ray μ CT, multi- and hyper-spectral cameras, isotope tracing methods). These techniques have recently been reviewed in detail [8,71] and can be used at a variety of scales to support the 4D functional analysis of root or shoot systems, and capture the structure and physiological status of plants (Figure 3). However, imaging devices and protocols perform photography, not phenotyping: traits need to be recovered from raw image data via image analysis (Figure 5, arrows 1 and 2). We discuss some of the key issues and solutions below.

Many software tools dedicated to image analysis of shoots [72], roots [73,74], canopies [75], leaves [76], seeds [77] and fruit [78] have been developed in recent years. An increasing number of these tools offer realistic and non-invasive 3D reconstructions of plant organs [79], based on the combination of multi-view

Figure 2. Plant root phenotyping pipeline using X-ray micro-computed tomography (μ CT).

(A) μ CT scanning system to non-invasively image columns of soil-grown plants (ranging in resolution from 0.5–150 μm). (B) Example 2D cross-sectional image generated with μ CT scanner showing root material (in red) and the heterogeneous structure of soil (soil and water in grey, air spaces in black). (C) Image analysis software [74] can be used to recover root system of maize from the μ CT volume data after segmenting roots from thousands of 2D image slices, (D) and quantify root system traits, (E) to discover new root responses to environmental signal, like how soil water distribution patterns the positioning of lateral root branches [23], and (F) parameterise models to simulate growth and foraging for natural resources by root systems.

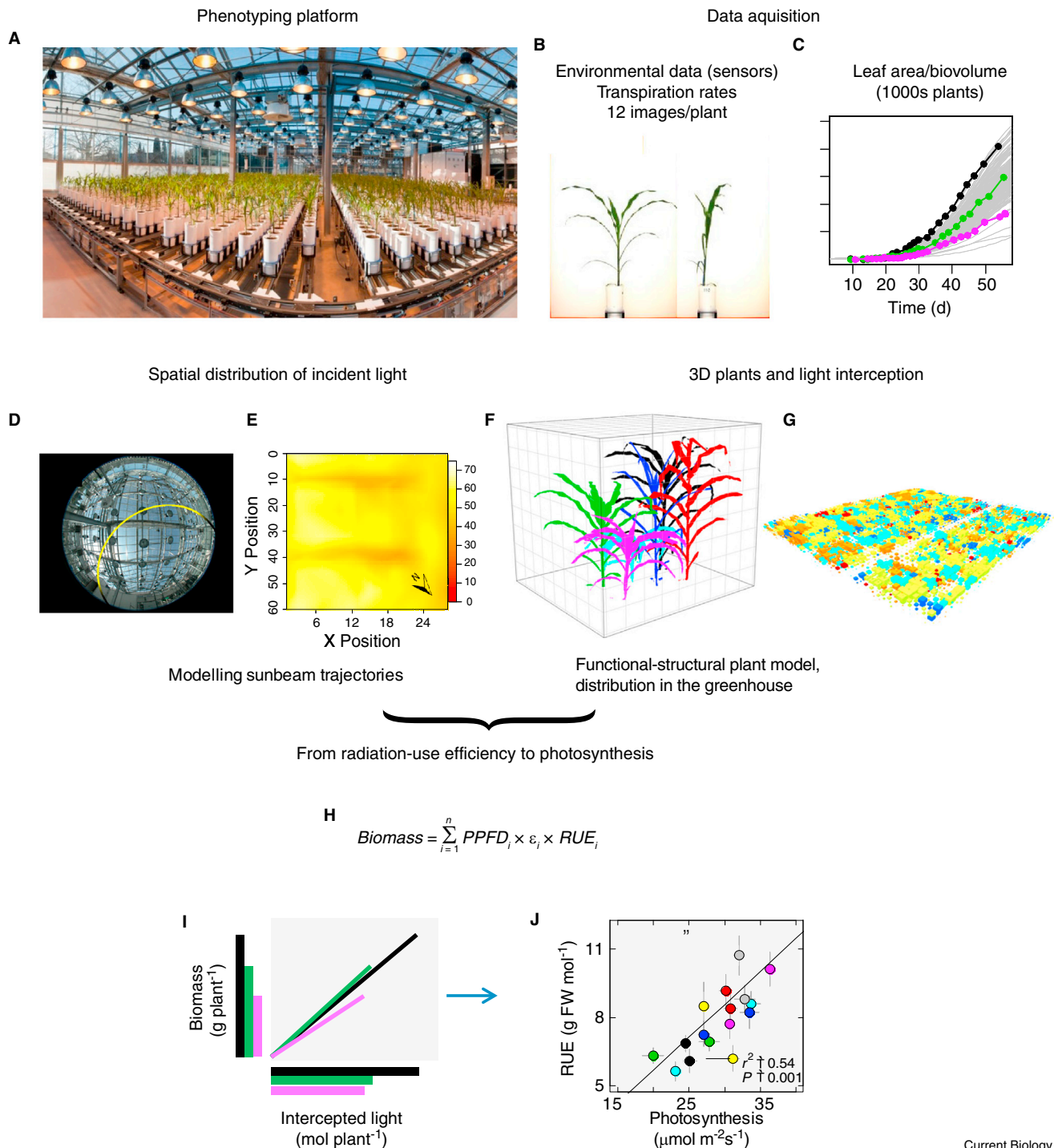


Figure 4. Light interception, photosynthesis and radiation use efficiency, from images to function.

(A) Phenotyping platform (PhenoArch) where 1680 plants can be grown in controlled conditions of soil water status and temperature, imaged and assessed for transpiration rate. Sensors measure light, relative humidity and air and leaf temperature and transpiration. (B) Twelve images per plant are captured every day allowing 3D reconstruction. (C) Time courses of leaf area and biovolume are calculated in real time. (D) Spatial distribution of incident light. Images are captured every m^2 in the greenhouse, oriented to the vertical. Blue, sky; black, obstacles (lamps, beams, etc.). The path of sunbeams is modelled every day of the year (yellow line). This allows calculation of direct and diffuse light in every position of the greenhouse [58]. (F) Virtual digital plants are placed at their positions in a virtual greenhouse. (G) This allows calculation of light interception by competing plants, in the whole greenhouse [58]. (H) The above steps allow dissection of biomass accumulation into incident light on day i (PPFD_i), the proportion of light intercepted by plants (ϵ_i) and radiation use efficiency (RUE_i , ratio of biomass production to intercepted light). (I) RUE is presented for three plants in (F), pink, green and black. Bars near the x and y axes represent the amounts of cumulated biomass and intercepted light, the slope of regression lines is RUE. (J) RUE closely correlates with photosynthesis rate in a series of genotypes denoted by different colors. Note that it would be impossible to directly measure gas exchanges for 1680 plants. Adapted from [58] with permission.

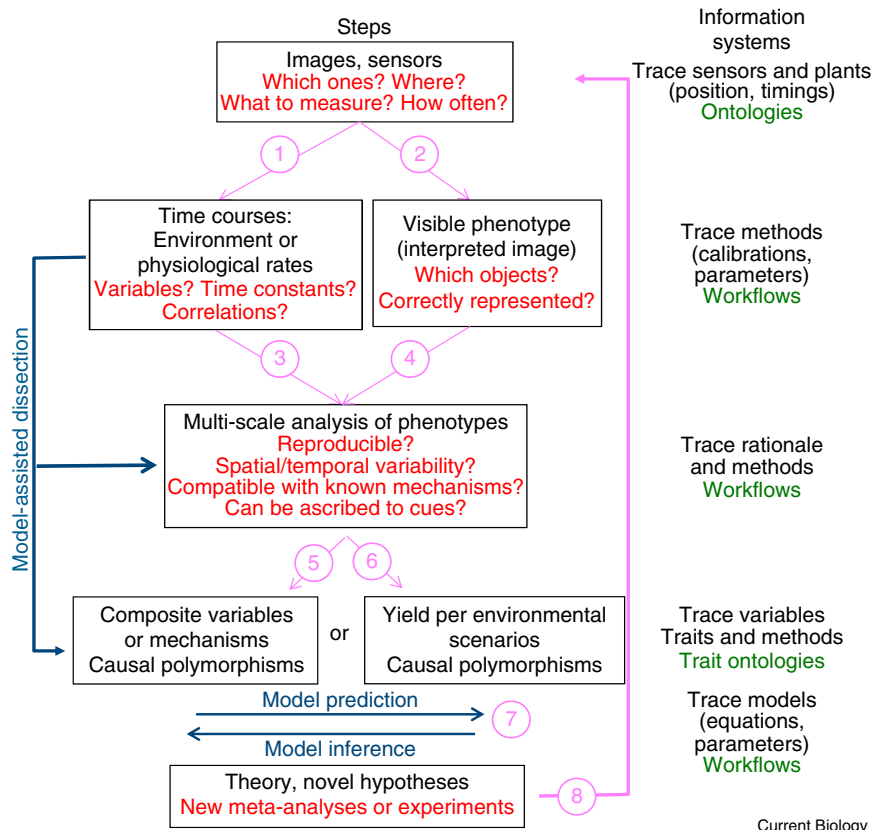


Figure 5. Flow chart of operations during phenotyping; roles of information systems and modelling.

The left panel represents steps from image/sensor to knowledge; the right panel represents the rationale for information systems at each step (green: tools). Red text represents questions at each step. Dark blue arrows and text: modelling tools. Purple arrows: connection between steps. (1) Transforming raw data into time courses for environmental data, fluxes, growth rates etc. (2) Image analysis to transform a series of images into a phenotype. (3) and (4) Data analysis with statistical and modelling tools, reproducibility. (5) Extraction of mechanisms or composite variables encapsulating the genotype x environment interaction, genetic analysis (6) association of yields to environmental scenarios, genetic analysis. (7) Prediction and inference of mechanisms vs scenario-dependent yields using models. (8) Theory, test using meta-analysis and/or new experiments.

images are acquired. This was done with success in a multi-laboratory study using *Arabidopsis thaliana* accessions grown in controlled chambers [91], but requires further attention. A phenotyping platform might give different assessments of the same genotype at two different sites, either because of environmental changes or as the result of variations in the phenotyping process. Image analysis methods need to be

stereo [80] and modelling [81,82], or use laser-scanning systems [83,84], time-of-flight sensors [85,86], X-ray [74,87] or magnetic resonance imaging [88]. Because plants are structurally complex and highly variable, a given set of sensor or camera viewpoints at fixed positions cannot provide all the data needed to reconstruct a complete 3D model of a plant or a canopy. Partial descriptions recovered from an initial set of camera views can be used, by solving a next-best-view problem, to guide a robot to acquire the data needed to complete the model. Indeed, robot-assisted imaging allows a loop to be established between image acquisition, analysis and *de novo* positioning of sensors at the most insightful places in plants [61,83,89]. This opens the way for a dialogue between models, sensors and imaging, enabling high-throughput, high-performance phenotyping of plants or canopies.

Interpretation of sensor or camera outputs requires the millions of raw data points to be organized into environmental or phenotypic time courses. This first requires the identification of dubious points due to sensor malfunction or computational errors, inevitable when thousands of sensors are involved, or when thousands of images are automatically processed. Such data cleaning can now be performed based on statistical or machine-learning methods for the large datasets originating from high-throughput platforms [90].

Data Analysis and Reproducibility Tests

Making reproducible measurements of the same plants or accessions over time and across platforms requires standardized protocols, including camera calibration, careful selection of number and position of viewpoints and the time of day at which

both understood [92] and evaluated by comparing their results with pre-obtained ground truth data [93], allowing identification of the limitation of each method [94].

The wealth of methods used in phenomics (Figures 3 and 4) raises the question of how to jointly analyse image and sensor outputs (Figure 5, arrows 3 and 4). Mixed model approaches have progressed rapidly, allowing genetic analysis of datasets involving different sources of information [95,96]. Novel developments allow identification of genotypic means of any variable, from omics to yield, which are isolated from the noise created by the spatial variability in field or platform experiments (Figures 3C,F and 4E), the effect of experimental co-variables (e.g., site, or persons who performed experiments) and environmental variables [97]. These 'best linear unbiased estimates' (BLUEs) are then analysed genetically either for individual traits or in multi-trait analyses [98].

Model-Assisted Phenotyping: Connecting Scales

Models naturally partner with phenotyping (Figure 5, arrow 7). For example, dynamic models offer the possibility of scaling up the effects of a short-term mechanism at the organ scale, identified in 'high-precision platforms', to biomass accumulation after several time steps in 'whole-plant, multi-environment platforms', or to yield in field networks. Dynamic models are based on the discretization of a process into time steps (e.g., minutes or days). Calculations are iterative, with short-term effects taken into account at each time step (e.g., the effect of light on photosynthesis, with different effects between genotypes), and long-term effects emerging from feedbacks

(e.g., the uptake of water or nutrients by the plant at a given time step reduces their availability for the next time step) [99]. Models have been used in plant phenomics in two ways [60].

Firstly, the dissection of a phenotype observed on a given day into the most likely set of mechanisms (model inference; Figure 5, arrows 5 and 7). For example, the biomass on a given day can be dissected into the amount of light received by the plant, multiplied by the proportion of light intercepted by plants every day, multiplied by the efficiency with which intercepted light is converted into biomass (Figure 4). Similarly, leaf area can be analysed as the result of time courses of leaf growth over time, resulting from environmental conditions and intrinsic traits of the considered genotype [100].

Secondly, the prediction of a given phenotype from environmental conditions and hypothetical mechanisms observed in high-precision or whole-plant platforms (Figure 5, arrow 7). Model prediction operates in the opposite direction compared with dissection, and serves as a test for the proposed mechanisms based on their ability to account for an observed phenotype. The set of mechanisms taken into account are written as equations which result in a phenotype after several time-steps [101].

Hence, modelling is an essential tool for phenomics because it helps to develop hypotheses allowing multi-scale interpretations of results obtained in the three types of phenotyping infrastructures presented in Table 1. Reciprocally, multi-scale phenomics represents a major challenge for modelling. Indeed, phenomic technology allows multiple traits that contribute to yield to be measured at high temporal resolution, providing a rich data set against which models can be tested [101]. This avoids compensation of errors associated with each trait underlying yield, a common feature of many current crop models that are parametrized based on yield only [102].

Tracing and Storing All Steps from Data to Knowledge in Information Systems

Phenomic experiments are not directly reproducible because of the variability of environmental conditions. It is essential that any scientist, including those in 30 years, be able to re-use phenotypic data and reproduce the data-flows presented above to perform meta-analyses of the effects of alleles or mechanisms in a range of environmental conditions. This has led to the definition of new norms named FAIR (findable, accessible, interoperable and reusable) [103], primarily for tracing data, but also protocols, methods and workflows. They involve information systems capable of managing thousands of data points and images captured during an experiment, together with the necessary metadata, parameters and methods of data analysis (Figure 5). Such information systems serve three distinct purposes with different requirements [104–107].

The first purpose is real-time management of the dataflow to optimise data quality. Real-time access to images, environmental conditions and metadata is required when managing the quality of an experiment, in particular for testing (typically every day) the validity of outputs. This may seem trivial in small-scale experiments but it is not when thousands of plants and hundreds of sensors are involved. Protocols [108,109] and management tools [90] have been developed to visualize large

volumes of temporal data in real-time, thereby allowing one to detect potentially incorrect sensors and to act accordingly.

Secondly, these information systems help organize datasets in such a way that they can be re-analysed by different groups. Data identification and annotation involves organizing outputs in such a way that a scientist not involved in the original experiments can trace the history of plants, re-analyse images with new methods of his/her own and *a posteriori* check the calibration of each sensor in case of inconsistencies, possibly years after the experiment was performed.

This requires protocols describing content and format of phenotypic information [110], and a formalised description of all involved objects (i.e., plants, organs, sensors, phenotyping facilities) using ontologies [111,112]. Such ontologies may seem unnecessary in simple experiments where unique correspondences exist between, for example, each plant and its position in a greenhouse. They become indispensable, however, when plants are transferred from one platform to another during an experiment for better multi-scale characterization. In the same way, sensors are replaced, so calibrations of devices located at a given position change with time. Keeping track of these changes requires open and extensible database schemas based on ontologies and semantics [111]. This also requires keeping track of all operations, including parameters, used in analyses that produce an elaborate result from raw data. Such scientific workflows are being developed [110], thereby allowing any user to perform the same analysis and obtain the same results as those published.

Finally, these systems help organise data to facilitate genetic analyses. Correspondence between phenotype and genotype requires connection of matrices of genotypic data, consisting of millions of marker data items or genomic sequences, with associated phenotypic data that synthesize time courses or spatial variation into single figures supporting the genetic analyses [113]. Because of the complexity of the information systems reviewed above, and of the need for high calculation power, this is performed in dedicated information systems that are physically distinct from those managing dataflow and object identification. Hence, maintaining consistency of information across multiple information systems will remain a major issue.

The 'Big-Data' Challenge of Plant Phenomics

Big data approaches can enhance phenotyping pipelines. Image analysis methods have typically employed fixed sequences of image processing and measurement processes, crafted by their designers to suit specific procedures. As a result, moving a given tool to a slightly different problem or environment often requires a near-complete rewrite of the software. Recently, deep machine-learning methods, and particularly convolutional neural networks (CNNs), have produced impressive results and been widely adopted in the computer vision community [114,115]. CNNs offer the potential to provide generic solutions to plant image analysis problems [116] and, rather than requiring tuning to their environment, benefit most from access to training data spanning multiple environments. This brings its own challenges — maximum benefit can only be gained from deep-learned tools if large-scale datasets (input images and required outputs) capturing shared problems are made available.

Big data approaches may also help in making full use of tens of experiments with thousands of accessions carried out each year. The formalized meta-analysis of phenotypic data, allowed by the pipelines reviewed above, is critical to the pathway from sensors to knowledge, and would be a huge source of information if data were open, with all necessary metadata and environmental conditions included [117]. Indeed, the discussion above suggests that the combination of datasets collected by distinct groups from different phenotyping platforms and fields could result in unprecedented information that may build up year after year. Recent papers present ‘proofs of concept’ of the meta-analysis of large datasets combining environmental and phenotypic data [118–120], and discuss their role in multi-environment quantitative genetics [121].

Finally, the meta-analyses of tens of experiments may be combined with data collected by farmers on yield and environmental conditions, thereby allowing large-scale association genetics. The sensor networks that are appearing in farmers’ fields, multi-layer maps of climate and soil characteristics and progress in remote sensing may soon provide the environmental data necessary to interpret the diversity of yield corresponding to each variety in each field. If large-scale collections of yield and environmental conditions in farmer’s fields were organized, association genetics at the level of countries or continents would become possible. This type of approach is already operational in big-data analyses of, for example, human social media behaviour, and its adoption in phenomics is of interest to a range of stakeholders.

Concluding Remarks

Plant phenomics research faces a conceptual challenge. To date, researchers have focused on employing and/or developing novel sensors and imaging techniques [8–10]. However, methodological advances in terms of data acquisition, handling and processing are becoming increasingly important. Indeed, the challenges of translating sensor information into knowledge have been grossly underestimated during the first years of plant phenomics research. Facing this challenge involves taking into account the intimate interaction between environmental conditions and plant structure, functions and metabolism, which require environmental characterization to be part of all steps of phenotyping, from data collection to meta-analyses. It also requires the use of both dynamic and statistical models allowing multi-scale analyses across experiments and platforms, which are essential to deal with the plant peculiarities reviewed at the beginning of this paper. Finally, the most recent advances in information technology must be employed to face the big-data challenge associated with multi-image processing, of meta-analysis of heterogeneous data and of the deployment of phenomics beyond the strict world of research. For obvious budget issues, it will not be possible to monitor all temporal and organization scales in every environment, but we believe that the rapid progress in modelling and information systems will allow identification of adequate cocktails of equipment, methods and meta-analyses allowing optimization of resources.

Hence, we propose that phenomics has reached a stage at which the limiting step is the design of methods and approaches allowing one to take into account different temporal and spatial scales and perform meta-analyses for addressing the challenges

of plant adaptation to changing environments and underpin secure food security efforts.

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